



Stable sets in k -colorable P_5 -free graphs

Frédéric Maffray

C.N.R.S., Laboratoire G-SCOP, Grenoble, France

ARTICLE INFO

Article history:

Received 15 April 2009

Received in revised form 4 September 2009

Accepted 14 September 2009

Available online 16 September 2009

Communicated by B. Doerr

Keywords:

Graph algorithm

Stability

ABSTRACT

We show that, for fixed k , there is a polynomial-time algorithm that finds a maximum (or maximum-weight) stable set in any graph that belongs to the class of k -colorable P_5 -free graphs, or, more generally, to the class of P_5 -free graphs that contain no clique of size $k + 1$. This is based on the following structural result: every connected k -colorable P_5 -free graph has a vertex whose non-neighbors induce a $(k - 1)$ -colorable subgraph.

© 2009 Elsevier B.V. All rights reserved.

1. Introduction

In a graph G , a *stable set* (also called independent set) is any subset of pairwise non-adjacent vertices. The MAXIMUM STABLE SET PROBLEM (henceforth MSS) is the problem of finding a stable set of maximum size. In the weighted version of this problem, we are given a weight for each vertex of G , and the weight of any subset of vertices is defined as the total weight of its elements. The MAXIMUM WEIGHTED STABLE SET PROBLEM (MWSS) is then the problem of finding a stable set of maximum weight. MSS (and consequently MWSS) is NP-hard in general, even under strong restrictions [2,10].

Given a graph F , a graph G is said to be F -free if no induced subgraph of G is isomorphic to F . Say that a graph is of *type T* if it is a tree with no vertex of degree at least four and with at most one vertex of degree three. Alekseev [1] proved that MSS remains NP-complete in the class of F -free graphs whenever F is a graph of which at least one component is not of type T. On the other hand, there are connected graphs F of type T for which the complexity of MSS in the class of F -free graphs is not known. The path on five vertices P_5 is the smallest such graph. This has attracted the attention of many researchers and there are many results dealing with MSS in special subclasses of

P_5 -free graphs; see [3] for a survey with many references and [8] for a recent result.

In any graph G , a k -coloring is a mapping $c : V \rightarrow \{1, \dots, k\}$ such that any two adjacent vertices u, v satisfy $c(u) \neq c(v)$. A graph G is called k -colorable if it admits a k -coloring. Here we show that, for fixed k , there exists a polynomial-time algorithm that solves MSS (or the weighted version MWSS) in the class of k -colorable P_5 -free graphs.

For any vertex v in a graph G , let $N(v)$ denote the set of all neighbors of v .

2. Bipartite P_5 -free graphs

Recall that a graph G is *bipartite* if its vertex-set can be partitioned into two stable sets. In this case we write $G = (A, B; E)$, where A, B are two stable sets that form such a partition. The structure of bipartite P_5 -free graphs is completely elucidated in the following theorem.

Theorem 1. (See [9].) *Let $G = (A, B; E)$ be a connected bipartite P_5 -free graph with $E \neq \emptyset$. Then:*

- Any two vertices u, v of A satisfy either $N(u) \subseteq N(v)$ or $N(v) \subseteq N(u)$, and the same holds for B .
- There is an integer $h \geq 1$, a partition $A_1, \dots, A_h$ of A into h non-empty sets and a partition $B_1, \dots, B_h$ of B into h non-

E-mail address: frederic.maffray@g-scop.grenoble-inp.fr.

empty sets such that a vertex in A_i and a vertex in B_j are adjacent if and only if $i + j \leq h + 1$.

- G has exactly $h + 1$ maximal stable sets, which are A, B and $A_{i+1} \cup \dots \cup A_h \cup B_{h+1-i} \cup \dots \cup B_h$ for each $i = 1, \dots, h - 1$.

Note that, with the same hypothesis and notation as in Theorem 1, every vertex of A_1 is adjacent to all vertices of B , and every vertex of B_1 is adjacent to all vertices of A . Thus, for every vertex x of $A_1 \cup B_1$, the non-neighbors of x induce a stable set. We will show in the next theorem that a similar kind of property holds in every connected P_5 -free graph. From an algorithmic point of view, note that the vertices of A_1 have degree $|B|$, whereas the vertices of $A \setminus A_1$ have degree at most $|B \setminus B_h|$. So the vertices of A_1 are precisely the vertices of largest degree among all vertices of A ; and the same holds in B .

3. Special vertices in P_5 -free graphs

Theorem 2. *Let G be a connected k -colorable P_5 -free graph with $k \geq 2$. Then G has a vertex whose non-neighbors induce a $(k - 1)$ -colorable subgraph. Moreover, G has at most n^{k-1} maximal stable sets, where n is the number of vertices of G .*

Proof. We prove the theorem by induction on n . If $k = 2$, the result follows from Theorem 1, as remarked above. Now assume $k \geq 3$. Consider a k -coloring of the vertices of G , that is, a partition of $V(G)$ into k stable sets $S_1, \dots, S_k$. Consider the graph $G' = G \setminus S_1$. So G' is a $(k - 1)$ -colorable graph.

First suppose that G' is connected. By the induction hypothesis, G' has a vertex x whose set X' of non-neighbors in G' induces a $(k - 2)$ -colorable subgraph. The set of non-neighbors of x in G is a subset of $X' \cup S_1$, which induces a $(k - 1)$ -colorable subgraph, so x is the desired vertex of G .

Now suppose that G' is not connected. Let $B_1, \dots, B_t$ ($t \geq 2$) be the components of G' . Let H be the graph obtained from G by contracting each component of G' into one vertex, in other words, each B_i is replaced by a vertex b_i and there is an edge $b_i x$ in H if and only if $x \in S_1$ and there is an edge ux in G for some $u \in B_i$. Put $B = \{b_1, \dots, b_t\}$. So H is a bipartite graph with bipartition (S_1, B) . Since G is connected, H is connected. We observe that:

H is P_5 -free. (1)

For suppose on the contrary that H contains an induced path $u-v-w-x-y$. First suppose that u, w, y are in B and v, x are in S_1 . Let $u = b_1$, $w = b_2$, $y = b_3$. By the definition of H , there exist a vertex $u_1 \in B_1$ adjacent to v , a vertex $y_3 \in B_3$ adjacent to x , and vertices $w_2, w'_2 \in B_2$ such that w_2 is adjacent to v and w'_2 is adjacent to x . Since B_2 is connected, there is a chordless path P between w_2 and w'_2 in B_2 , and we may choose w_2, w'_2 so that P is as short as possible (possibly $w_2 = w'_2$). But then $u_1-v-w_2-P-w'_2-x-y_3$ is a chordless path with at least five vertices in G , a contradiction. When u, w, y are in S_1 and v, x are in B , the argument is similar and we omit the details. Thus (1) holds.

Since H is a connected P_5 -free bipartite graph, by Theorem 1 there is a vertex z of S_1 that is adjacent in H to all vertices of B . Thus, in G , vertex z has a neighbor in each component B_i ($i = 1, \dots, t$). Suppose that z has a non-neighbor in at least two of the components $B_1, \dots, B_t$, say in B_1 and B_2 . Since B_1 is connected, there are adjacent vertices u_1, v_1 in B_1 such that z is adjacent to u_1 and not to v_1 , and similarly there are adjacent vertices u_2, v_2 in B_2 such that z is adjacent to u_2 and not to v_2 . But then $v_1-u_1-z-u_2-v_2$ is an induced P_5 in G , a contradiction. Therefore we may assume, up to symmetry, that z is adjacent in G to all vertices of $B_2 \cup \dots \cup B_t$. Let $N_1 = N(z) \cap B_1$ and $M_1 = B_1 \setminus N(z)$. We know that $N_1 \neq \emptyset$. We claim that:

M_1 induces a $(k - 2)$ -colorable subgraph of G . (2)

For suppose the contrary. So there is a component M of the subgraph of G induced by M_1 that is not $(k - 2)$ -colorable. Since B_1 is connected and $N_1 \neq \emptyset$, there is a vertex u in N_1 that has a neighbor in M . If u has also a non-neighbor in M , then, since M is connected, there are adjacent vertices x, y in M such that u is adjacent to x and not to y ; and then $y-x-u-z-w$ is an induced P_5 in G , where w is any vertex in B_2 , a contradiction. Thus u is adjacent to all vertices of M . But then the subgraph induced by $M \cup \{u\}$ is not $(k - 1)$ -colorable, which contradicts the fact that it is a subgraph of G' . Thus (2) holds.

Since the set of non-neighbors of z in G is $M_1 \cup S_1 \setminus \{z\}$, (2) implies that z is the desired vertex of G .

To complete the proof of the theorem, let us count the maximal stable sets of G . By the induction hypothesis, the number of maximal stable sets that contain z is at most $(n - 1)^{k-2}$, and the number of those that do not contain z is at most $(n - 1)^{k-1}$. Thus the total number of maximal stable sets of G is at most $(n - 1)^{k-1} + (n - 1)^{k-2} \leq n^{k-1}$. $\square$

The proof of Theorem 2 can easily be seen as a polynomial-time algorithm FIND_k which, given a connected P_5 -free graph G with n vertices and a k -coloring $S_1, \dots, S_k$ of G , finds the desired vertex. Algorithm FIND_k works as follows:

- If $k = 2$, return any vertex of largest degree. (This takes time $O(n^2)$.)
- If $k \geq 3$ and $G \setminus S_1$ is connected, then apply FIND_{k-1} on $G \setminus S_1$.
- If $k \geq 3$ and $G \setminus S_1$ is not connected, construct the graph H (this takes time $O(n^2)$), and return a vertex of S_1 of largest degree in H .

Because the second step may be repeated $k - 2$ times when $k \geq 3$, the overall complexity of this algorithm is $O(n^3)$ when $k \geq 3$ and $O(n^2)$ when $k = 2$.

We point out two differences between Theorems 1 and 2:

- In Theorem 1 ($k = 2$), every color class contains a vertex whose non-neighbors induce a $(k - 1)$ -colorable subgraph. This is not true for non-bipartite graphs. For example, let G be the graph that consists of two

disjoint cliques X, Y of size $k \geq 3$ plus an edge xy between one vertex x of X and one vertex y of Y . Clearly, G is P_5 -free. Consider any coloring of G with k colors and assume that x has color 1 and y has color 2. Then color classes $3, \dots, k$ do not contain any vertex whose non-neighbors induce a $(k-1)$ -colorable subgraph.

- (ii) In Theorem 1 ($k=2$), any vertex of largest degree in its class is a vertex whose non-neighbors induce a $(k-1)$ -colorable subgraph. This is not true for non-bipartite graphs, even when the class does contain such a vertex. For example, let G be the graph with vertices $u, v, w, x, y, z_1, z_2, z_3$ and edges $uv, uw, vw, wx, xy, z_i x, z_i y$ ($i=1, 2, 3$). Clearly, G is P_5 -free. Consider the coloring of G such that w, y have color 1, u, x have color 2, and v, z_1, z_2, z_3 have color 3. Then y is the vertex with largest degree in color class 1, yet y is not a vertex whose non-neighbors induce a 2-colorable subgraph (while w is such a vertex).

These two remarks explain why, in Algorithm FIND_k , we cannot apply the same procedure when $k \geq 3$ as when $k=2$ and we must use recursion instead.

Recall that the *chromatic number* of a graph G is the smallest k such that G admits a k -coloring. Determining the chromatic number of a P_5 -free graph is NP-hard [7]. However, Hoàng, Kamiński, Lozin, Sawada and Shu [6] gave a polynomial-time algorithm that, for fixed k , determines if a P_5 -free graph is k -colorable (and if it is, returns a k -coloring). This algorithm can be used before applying our algorithm FIND_k in case a k -coloring was not given as input of our algorithm. When $k=3$, a polynomial-time algorithm that decides whether a P_5 -free graph is 3-colorable was also given by Woeginger and Sgall [12] and by Randerath and Schiermeyer [11].

Theorem 3. For fixed $k \geq 1$, there exists an $O(n^k)$ algorithm that finds a maximum weighted stable set in any k -colorable P_5 -free graph.

Proof. We prove the theorem by induction on k . Here is an algorithm A_k that solves this problem. Let $G=(V, E)$ be a k -colorable P_5 -graph with a weight function on its vertices. For $k=1$, simply return the set V . Now assume $k \geq 2$.

- If G is not connected, solve the problem for each component of G and merge the solutions to get a solution for G .
- If G is connected, then, by Theorem 2, G has a vertex x whose non-neighbors induce a $(k-1)$ -colorable subgraph G_x . With algorithm FIND_k we can find such a vertex in time $O(n^3)$ if $k \geq 3$ and $O(n^2)$ if $k=2$. Then proceed as follows:
 1. Find a maximum weighted stable set S_x in G_x , using Algorithm A_{k-1} .
 2. Find a maximum weighted stable set S' in $G \setminus x$ using A_k .
 3. Among the two stable sets $S_x \cup \{x\}$ and S' , return the set with larger weight.

Clearly, for any vertex x , a maximum weighted stable set S of G either contains x or not. If it contains x , then $S \setminus \{x\}$ is a maximum weighted stable set of G_x . If it does not contain x , then it is a maximum weighted stable set of $G \setminus x$. This ensures the correctness of algorithm A_k .

By the induction hypothesis, Step 1 takes time $O(n^{k-1})$. The recursive Step 2 is called at most n times. So the total complexity of algorithm A_k is $O(n^k)$. $\square$

4. P_5 -free graphs with bounded clique size

For fixed k , let us consider the class $\mathcal{C}_k$ of P_5 -free graphs that do not contain a clique of size $k+1$; this class clearly extends the class of k -colorable P_5 -free graphs. It was proved in [5] that, for each $k > 0$, there exists a constant c_k (which depends only on k) such that every graph in $\mathcal{C}_k$ is c_k -colorable. Hence, it follows from Theorem 3 that, for fixed $k \geq 2$, there exists a polynomial-time algorithm, of complexity $O(n^{c_k})$, that finds a maximum weighted stable set in any graph in $\mathcal{C}_k$ with n vertices. The value obtained in [5] is $c_k \leq 4^{k-1}$. This was improved in [4] to $c_k \leq 3^{k-1}$.

5. Conclusion

We proved that for fixed k , there is an algorithm that solves MWSS in polynomial time in the class of k -colorable P_5 -free graphs, or, more generally, in the class of P_5 -free graphs that do not contain a clique of size $k+1$. Since P_5 -free graphs can have arbitrarily large chromatic number and clique size, this does not yield a polynomial-time algorithm for MSS in the general class of P_5 -free graphs. The complexity status of MSS in P_5 -free graphs remains open.

References

- [1] V.E. Alekseev, On the local restrictions effect on the complexity of finding the graph independence number, in: Combinatorial Algebraic Methods in Applied Mathematics, Gorkiy Univ. Press, 1983, pp. 3–13 (in Russian).
- [2] M.R. Garey, D.S. Johnson, Computers and Intractability: A Guide to the Theory of NP-Completeness, W.H. Freeman, 1979.
- [3] M. Gerber, V. Lozin, On the stable set problem in special P_5 -free graphs, Discrete Applied Mathematics 125 (2003) 215–224.
- [4] S. Gravier, C.T. Hoàng, F. Maffray, Coloring the hypergraph of maximal cliques of a graph with no long path, Discrete Mathematics 272 (2003) 285–290.
- [5] A. Gyárfás, Problems from the world surrounding perfect graphs, in: Proc. International Conf. on Combinatorial Analysis and its Applications, Pokrzywna, 1985, Zastosowania Matematyki XIX (1987) 413–441.
- [6] C.T. Hoàng, M. Kamiński, V. Lozin, J. Sawada, X. Shu, A note on k -colorability of P_5 -free graphs, Lectures Notes in Computer Science 5162 (2008) 387–394.
- [7] D. Král', J. Kratochvíl, Zs. Tuza, G.J. Woeginger, Complexity of coloring graphs without forbidden induced subgraphs, Lectures Notes in Computer Science 2204 (2001) 254–262.
- [8] V. Lozin, R. Mosca, Maximum independent sets in subclasses of P_5 -free graphs, Information Processing Letters 109 (2009) 319–324.
- [9] N.V.R. Mahadev, U.N. Peled, Threshold Graphs and Related Topics, Annals of Discrete Mathematics, vol. 56, Elsevier, Amsterdam, 1995.
- [10] O. Murphy, Computing independent sets in graphs with large girth, Discrete Applied Mathematics 35 (1992) 167–170.
- [11] B. Randerath, I. Schiermeyer, 3-Colorability in P for P_6 -free graphs, Discrete Applied Mathematics 136 (2004) 299–313.
- [12] G.J. Woeginger, J. Sgall, The complexity of coloring graphs without long induced paths, Acta Cybernetica 15 (2001) 107–117.